AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

M218
Relational Databases

Midterm Examination (MTA) - Makeup
Spring Semester 2024/2025

Date: 22/April/2025

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed:** 1.5 Hours

Instructions:

- Total Marks: **30**
- This exam consists of **2** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name** and **ID** on the External Answer booklet.
- **Calculators** are allowed.

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Part 1: Multiple choice questions [5 Marks]

This part consists of 5 questions carrying a weight of 1 mark for each. Answer the questions by choosing the most correct choice.

1. Which of the following is a key advantage of the Relational Model over the Network and Hierarchical Models?
    a. It uses a complex pointer-based navigation system
    b. It simplifies data retrieval using structured queries (SQL)
    c. It does not require a database schema
    d. It does not support relationships between tables

2. In the Three-Schema Architecture, what is the role of the Conceptual Level?
    a. It describes the physical storage details of the database
    b. It defines how data is organized for a community of users
    c. It provides individual user views by filtering database contents
    d. It represents the current snapshot of the database at a given moment

3. Which of the following is TRUE regarding the difference between a Database Schema and a Database State?
    a. A database schema changes frequently as new data is inserted or deleted
    b. A database state refers to the description of the database structure
    c. A database schema is fixed, while a database state changes over time
    d. Schema evolution means adding new data to an existing database

4. What is a relation in the relational model?
    a. A function that maps one attribute to another
    b. A set of tuples containing related data values
    c. A database stored as a single text file
    d. A network of connected entities

5. Which of the following statements about foreign keys and referential integrity is TRUE?
    a. A foreign key in a relation must always reference a candidate key in another relation
    b. Referential integrity ensures that foreign key values always match primary key values in the referenced table
    c. A foreign key constraint allows referencing an attribute that is not a primary key
    d. If a tuple is deleted, referential integrity is always violated

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Part 2: Problem solving questions [25 Marks]

Question 1 [8 Marks]

A Movie Industry Database is a system designed to manage information about movies, actors, and studios.

Entities and Attributes:

- Song (Song_ID , Title, Duration, Genre)
- Artist (Artist_ID , Name, DOB (Date of Birth), Nationality)
- Album (Album_ID, Title, Release_Year, Producer)

Relationships:

- An album contains multiple songs, but each song belongs to only one album.
- An artist can perform multiple songs, and a song can be performed by multiple artists.

Instructions:

1. Identify the primary keys and the foreign keys.
2. Draw an ER diagram incorporating these entities and relationships, ensuring you denote primary keys (PK) and foreign keys (FK) accordingly.

Question 2 [17 Marks]

A company wants to track employee shifts using the following unnormalized table:

EmployeeID	Employee Name	Department	Shift Date	Shift Start	Shift End	Supervisor Name	Supervisor Phone
201	Alice Johnson	HR	2024-03-01	08:00 AM	04:00 PM	Mark Smith	555-6789
201	Alice Johnson	HR	2024-03-02	08:00 AM	04:00 PM	Mark Smith	555-6789
202	Bob Williams	IT	2024-03-01	09:00 AM	05:00 PM	Lisa Green	555-1234
203	Charlie Evans	Finance	2024-03-01	10:00 AM	06:00 PM	Tom Brown	555-4321
203	Charlie Evans	Finance	2024-03-02	10:00 AM	06:00 PM	Tom Brown	555-4321

1. Identify the functional dependencies (FDs) in the table. **[3 points]**
2. Identify the problems with this design (e.g., redundancy, anomalies). **[4 points]**
3. Convert the table to First Normal Form (1NF). **[2 points]**
4. Convert the table to Second Normal Form (2NF). **[4 points]**
5. Convert the table to Third Normal Form (3NF). **[4 points]**

End of questions

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